Then the sum of the ranks can be easily computed as follows:

$$T = \sum_{i=1}^{n} r_i \cdot W_i$$

- If the underlying algorithms are identical the actual results from each sample will be evenly distributed throughout the combined grouping. If one algorithm provides better performance results its sample should be concentrated around the lower cost rankings.
- In the situation where the null hypothesis is true the expected rank and variance of T are:

$$E(T) = \frac{m \cdot (m + n + 1)}{2}$$

$$V(T) = \frac{mn \cdot (m + n + 1)}{12}$$

As with the Wilcoxon signed rank test, it can be shown that as $n, m \rightarrow \infty$ the distribution of T converges to a normal distribution. This property allows us to test the hypothesis that there is no difference between broker VWAP algorithms using the standard normal distribution with the following test statistic:

$$Z = \frac{T - E(T)}{\sqrt{V(T)}}$$

Comparison to Critical Value:

- Reject the null hypothesis H_0 if $|Z| > C_{\alpha/2}$.
- $C_{\alpha/2}$ is the critical value on the standard normal curve corresponding to the $1 - \alpha$ confidence level.
- For example, for a 95% confidence test (i.e. $\alpha = 0.05$) we reject the null hypothesis if $|Z| > 1.96$. Notice here we are only testing if the distributions are different (therefore a two tail-test).
- The hypothesis can also be constructed to determine if A has better (or worse) performance than B by specifying a one-tail test. This requires different critical values.

Analysts need to categorize results based on price trends, capitalization, side, etc. in order to determine if one set of algorithms performs better or worse for certain market conditions or situations. Many times a grouping of results may not uncover any difference.

An extension of the Mann Whitney U test used to compare multiple algorithms simultaneously is the Kruskal-Wallis one way analysis of